

Recent developments in ATDEP

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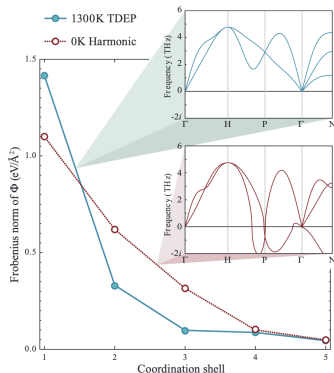
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Temperature-dependent effective potential (TDEP) method

■ Original TDEP formulation

O. Hellman et al. (2013). *Physical Review B* 87.10, p. 104111



■ a-TDEP: implementation in Abinit

F. Bottin et al. (2020). *Computer Physics Communications* 254, p. 107301



Temperature-dependent effective potential (TDEP) method

Expand the forces in powers of atomic displacements and interatomic force constants (IFC)

$$F_{i_1}^{\alpha_1} = - \sum_{p \geq 2} \frac{1}{(p-1)!} \sum_{\substack{\alpha_2 \dots \alpha_p \\ i_2 \dots i_p}} \Theta_{i_1 i_2 \dots i_p}^{\alpha_1 \alpha_2 \dots \alpha_p} \prod_{k=2}^p (\mathbf{R} - \mathbf{R}_0)_{i_k}^{\alpha_k}$$

Running a molecular dynamics (MD) produces a set of forces and atomic positions that define a linear equation for the IFC

$$F_{\text{MD}}^{\alpha,i}(t) = \sum_{p \geq 2} \sum_{\lambda} f_{p\lambda}^{\alpha,i}(\mathbf{R}(t)) \Theta_{p\lambda}$$

IFC are found by a least square minimization

$$\min_{\Theta} \|\mathbf{F}_{MD} - \mathbf{f} \cdot \Theta\| \longrightarrow \Theta$$

Requires supercell sampling with small atomic displacements.

Changes to the ATDEP interface

- **Standardized input file format**

Same input file syntax as other abinit executables (abinit, anaddb).

- **Command line interface**

Files file no longer needed. Standard input variables for files specification (`output_file`, `indata_prefix`, `outdata_prefix`)

- **Improved flexibility**

Accomodate arbitrary crystal orientation.

Allow for reconstructed `HIST.nc` files (e.g. from MLACS).

- **DDB reading**

Born effective charges and dielectric tensor can be read directly from a DDB file.

- **DDB output**

ATDEP now writes a DDB file, which can be processed by Anaddb.

Long-range forces

Non-analytic part of the dynamical matrix

$$\Theta_{j\alpha,j'\alpha'}^{NA}(\mathbf{q}) = \frac{4\pi e^2}{\Omega} \frac{(\mathbf{q} \cdot \mathbf{Z}_j^*)_{\alpha} (\mathbf{q} \cdot \mathbf{Z}_{j'}^*)_{\alpha'}}{\mathbf{q} \cdot \epsilon^{\infty} \cdot \mathbf{q}} \quad (1)$$

Long-range forces resulting from dipole-dipole interaction

$$F_{j\alpha}^{LR}(0) = - \sum_{\mathbf{R}',j',\alpha'} \Theta_{j\alpha,j'\alpha'}^{NA}(\mathbf{R}') u_{j'}^{\alpha'}(\mathbf{R}') \quad (2)$$

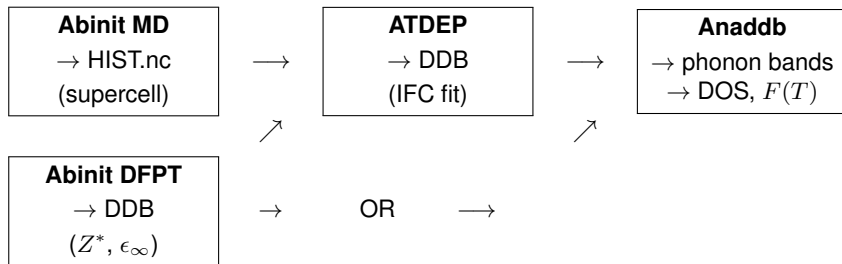
One obtains the short-range IFC by fitting it to the short-range forces

$$F_{j\alpha}^{SR}(\mathbf{R}) = F_{j\alpha}(\mathbf{R}) - F_{j\alpha}^{LR}(\mathbf{R}) \quad (3)$$

Then reconstruct

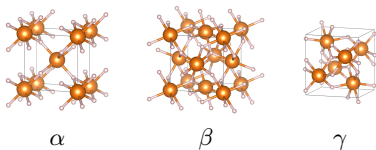
$$\Theta_{j\alpha,j'\alpha'}(\mathbf{R}') = \Theta_{j\alpha,j'\alpha'}^{SR}(\mathbf{R}') + \Theta_{j\alpha,j'\alpha'}^{NA}(\mathbf{R}') \quad (4)$$

Complete workflow for anharmonic phonons

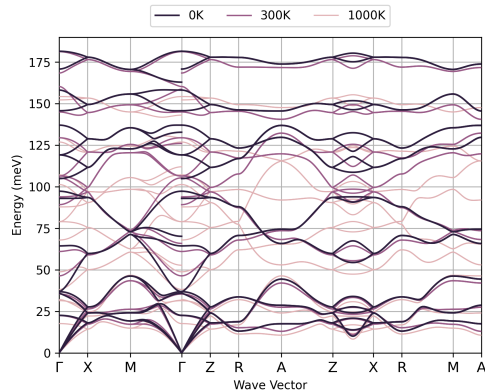
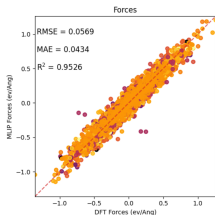


- AIMD calculation in a supercell commensurate with the q-points grid.
- DFPT calculation provides Z^* to ATDEP for long-range IFC contribution.
- Phonon bands calculation performed by Anadddb.

Phonon anharmonicity in MgH_2



- MD sampling with MLACS
- 20 DFT calculations
- 2000 MLIP-MD steps between each DFT



- Important phonon softening at $T > 300\text{K}$



- **AtdepInput** object.

- ▶ Initialized from a Structure object
- ▶ Automatic determination of Bravais lattice (brav)

- **AtdepTask** object

- ▶ Mandatory usage of HIST.nc file.
- ▶ Providing DDB activates the computation of long-range IFC.

Abipy interface

```
from pathlib import Path
import abipy.data as abidata
from abipy.abio.inputs import AtdepInput, AnaddbInput
from abipy import flowtk

def main():
    workdir = Path(__file__).name.replace(".py", "").replace("run_", "flow_")

    structure = abidata.structure_from_mpid('mp-1265') # MgO
    ddb_file = abidata.ref_file('MgO_dfpt_aimd/MgO_DDB.nc')
    hist_file = abidata.ref_file('MgO_dfpt_aimd/MgO_HIST.nc')

    flow = get_atdep_flow(workdir, structure, hist_file, ddb_file)
    flow.build_and_pickle_dump()
```

Abipy interface

```
def get_atdep_input(structure):  
  
    inp = AtdepInput(structure)  
    inp.set_vars(  
        multiplicity=[[-2,2,2],[2,-2,2],[2,2,-2]],  
        nstep_max=20,  
        nstep_min=1,  
        temperature=300,  
        rcut=7.0,  
        ngqpt1=[2,2,2],  
        ngqpt2=[1,1,1],  
    )  
  
    return inp
```

Abipy interface

```
def get_atdep_flow(workdir, structure, hist_file, ddb_file=None):
    atdep_inp = get_atdep_input(structure)
    anaddb_inp = get_anaddb_input(structure)

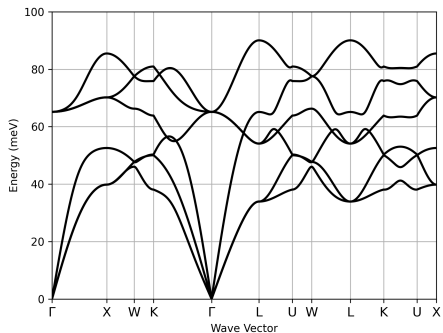
    anaddb_inp.set_vars(ngqpt = atdep_inp['ngqpt1'], rifcsph = atdep_inp['rcut'])

    atdep_task = flowtk.AtdepTask(atdep_inp, hist_node=hist_file, ddb_node=ddb_file)
    anaddb_task = flowtk.AnaddbTask(anaddb_inp, ddb_node=atdep_task)

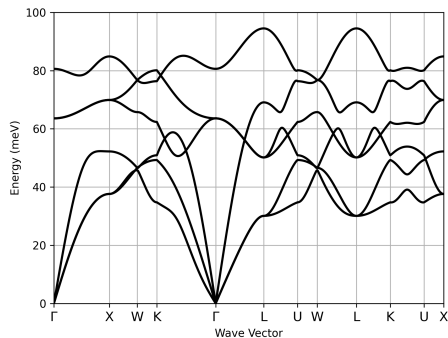
    flow = flowtk.Flow(workdir=workdir)
    flow.register_task(atdep_task)
    flow.register_task(anaddb_task, deps={atdep_task: "DDB"})
    return flow
```

Results for MgO example (low-parameters)

Without BEC



With BEC



Inclusion of BEC still under development

- May not be physically correct at the moment.
- Need to impose cut-off radius.

Improved atoms matching function

- Supercell definition should be independent atom displacements.
- Remove input variables: `tolmotif`, `tolinbox`, `tolmatch`,

Remove alloying

- ATDEP works the same with alchemical pseudoatoms.
- Remove `alloy` input variable.

IFC overhaul

- Module `m_ifc` is a mess!
- Several IFC file formats exist.
- Long-range forces only computed when IFC are written.

Remove `brav` input variable

- Present in both ATDEP and Anaddb.
- Space group should be detected internally.

Remove internal usage of `acell`

- `acell` is useful for specifying the unit cell.
- IFC definition should not depend on `acell`.
- Internally, we should always use `rprimd`.